PDTN Phase B Notes Study Plan

This file is not for learning new material.

It is for learning how to use the notes fast under pressure.

The goal for the last 2 evenings is:

- stop searching randomly
- know where each kind of problem lives in the notes
- find the right snippet quickly
- apply a baseline or fix without hesitation

1. Main Principle

Treat the notes like a tool, not a textbook.

Your goal is not:

- read everything again
- memorize every code block
- improve the notes again

Your goal is:

- memorize the map
- practice retrieval
- build pressure habits

2. The Map You Should Know By Memory

Memorize these sections first:

- `2` : first baseline by problem type
- `4` : sklearn / tabular fixes
- `6` : PyTorch errors, dtype, device, shapes, loss issues
- `7` : vision, transforms, channels, CNN reminders
- `8` : text, embeddings, TF-IDF, cosine similarity
- `9` : optimization / solver problems
- `10` : debugging checklist
- `11` : worked problems from this repo

If you know this map, you will already save time.

3. Fast Trigger -> Section Lookup

Use these trigger words during competition:

- `baseline` -> Section `2`

- `tabular / sklearn / NaN / encoding` -> Section `4`
- `dtype / device / shape / loss / DataLoader / batch` -> Section `6`
- `images / transforms / channels / pretrained backbone` -> Section `7`
- `TF-IDF / retrieval / cosine / embeddings / token ids` -> Section `8`
- `constraints / assignment / scheduling / discrete optimization` -> Section `9`
- `I do not know what is broken` -> Section `10`
- `this looks like an old solved repo problem` -> Section `11`

4. Competition Habit

When blocked, do this in order:

1. identify the task type
2. jump to the matching baseline section
3. implement the cheapest correct baseline
4. validate locally
5. if broken, jump to the matching error/debug section
6. only improve after a correct baseline exists

If you are stuck for more than 2 minutes:

1. classify the problem
2. open the matching section
3. copy the baseline pattern
4. use the debugging checklist

5. What To Practice

Do not just reread.

Practice lookup.

Use prompts like:

- tabular classification baseline?
- regression loss and target dtype?
- CrossEntropy target dtype?
- stack vs cat?
- binary classification loss?
- embedding retrieval normalization?
- DataLoader worker crash?
- submission sanity check?
- grayscale image into RGB backbone?
- coordinate-to-RGB regression like squarepainting?

For each prompt:

- start from the top of the notes
- find the correct section fast
- find the exact snippet or rule
- explain in one sentence why that section is the right one

6. Two-Evening Plan

Evening 1

Goal:

- memorize the structure of the notes

Steps:

1. skim the notes once for section structure only
2. write the section map from memory
3. review the section map until you can reproduce it
4. do 5 timed retrieval drills

Timed retrieval drill format:

- one prompt
- 30 to 60 seconds
- locate the correct section
- locate the relevant snippet
- say out loud what it is for

Evening 2

Goal:

- simulate contest retrieval under pressure

Use 3 to 5 mini-scenarios such as:

- image classifier not learning
- classification labels wrong dtype
- text task with no pretrained weights
- squarepainting-style RGB regression
- solver problem vs ML problem

For each scenario:

- start a timer
- use only the notes
- find the matching sections
- write or describe the first baseline or fix
- stop after 10 to 15 minutes

You are training navigation speed, not perfect solutions.

7. Five Emergency Queries To Memorize

These should become automatic:

- wrong task / wrong loss pairing
- shape mismatch
- device mismatch
- NaN / OOM
- submission / export mistake

If one of these appears, you should already know where to go.

8. What Not To Do

- do not passively reread the full notes again and again
- do not try to memorize every code block
- do not spend the last 2 evenings expanding the notes
- do not practice only when calm; use a timer
- do not let yourself browse aimlessly through sections

9. Best Mental Model

The notes are a decision system:

- identify task
- find baseline
- find error section if needed
- find similar worked problem if needed
- submit a correct version early

That is better than having bigger notes but no retrieval speed.

10. Five-Minute Pre-Competition Review

Right before the competition, review only this:

- the section map
- the trigger -> section mapping
- the 2-minute stuck rule
- the 5 emergency queries
- the baseline-first principle

Do not start deep studying at the last minute.

11. One-Line Rule

If I am stuck, I do not search randomly.

I classify the problem, jump to the matching section, use the cheapest correct baseline, and debug with the checklist.